1. DEFINITION

DEFINITION
Highly purified Diethylene glycol monoethyl ether EP/NF

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Ethoxydiglycol
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	A1A1I8X02B
ADDITIONAL INFORMATION ON THE DENOMINATION	DIETHYLENE GLYCOL MONOETHYL ETHER
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient It is a high purity excipient recommended for use in human pharmaceutical formulations administered by oral, topical and rectal/vaginal routes.

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Bulk DEGEE is obtained by condensation of ethylene oxide and alcohol (ethanol). Gattefossé purchases bulk DEGEE and performed a purification distillation in 2 steps in order to obtain an highly purified DEGEE.

COMPOSITION
Purified Diethylene glycol monoethyl ether (Specification: Purity > 99.8%)
Full Labeling (Key letters system) **$A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$**

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900

COMPOSITION	
Ethoxydiglycol	$A_1 \geq 75.0\%$
Other constituents (i.e. incidental ingredients)	
None	

4. DRUG MASTER FILE

Country	Master file number
US DMF (Type IV – Excipient)	5718
Canada MF (Type III – Excipient)	2005-108
For submission of the DMF in another country, please contact your local Gattefossé representative	

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	**Conforms to monograph**
European Pharmacopoeia Current edition	N°1198: Diethylene glycol monoethyl ether
USP/ NF Current edition	Diethylene glycol monoethyl ether Additional requirements: It is intended for topical or transdermal use only. The material is not to be used for parenterals.

CHINA BUNDLING REVIEW			
(Generic name) Chinese name	**Registration N°**	**Company**	**CFDA/CDE Reference** (http://www.cde.org.cn/yfb.do?method=main)
(Dietylene glycol monoethyl ether) 二乙二醇单乙醚	F20190000521	Gattefossé SAS	Raw material has not been approved yet for use in marketed formulations in Joint-Review with drug applications. However, the excipient dossier has already been submitted (Status I).
Please contact our Regulatory Affairs department for further information and obtain letter of authorization			

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900

6. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
◆	◆ (3) All ruminants, porcine (MRL: No limit)	◆	No known restriction	◆ (8)	◆ (8)

EUROPEAN REGULATION
(1) Regulation (EU) No 37/2010 - Listed in annex II (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption)
(2) Regulation (EU) No 37/2010 - Listed in annex II (Polyethylene glycol stearates with 8-40 oxyethylene units)
(3) Regulation (EU) No 37/2010 - Listed in annex II (Diethylene glycol monoethyl ether)
(4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin
USA REGULATION
- Veterinary drugs
FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume. In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety
(5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182—184 of the Code of Federal Regulations (CFR)
(6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database)
- Pesticides
(8)EPA's pesticides program (http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::)

7. SPECIFIC PHARMACEUTICAL REGULATIONS

CHINA BUNDLING REVIEW			
Generic name	Registration N°	Company	CFDA/CDE Reference (http://www.cde.org.cn/yfb.do?method=main)
Dietylene glycol monoethyl ether 二乙二醇单乙醚	F20190000521 (Status I) (Excipient)	Gattefossé SAS	Raw material approved for use in marketed formulations in Joint-Review with drug applications.
Please contact our Regulatory Affairs department for further information and obtain letter of authorization			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	Diethylene glycol monoethyl ether (ref 53203)	Available for use as an Active Ingredient in: Biologicals, Prescription Medicines. Available for use as an Excipient Ingredient in: Biologicals, Devices, Export Only, Listed Medicines, Over the Counter, Prescription Medicines. Not available as an Equivalent Ingredient in any application.	Listed Medicines: Only for use in topical medicines for dermal application
Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homoeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	**NHPID name**	**Route of administration/ Restrictions**
Approved non-medicinal ingredient	Ethoxydiglycol	Role Non-medicinal Purposes: Fragrance Ingredient , Solvent , Viscosity Decreasing Agent Route Of Administration: For topical use only. Role Non-NHP Rationale: A non-NHP because not a naturally occurring substance included in Schedule 1 of the NHP Regulations
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		

8. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSÉ					
Study type/ duration	**Route**	**Species**	**Test article**	**Results/Conclusions**	**Reference**
Percutaneous absorption (OECD 428)	*In vitro*	Human skin	C_{14}-TRANSCUTOL® Shampoo: 5, 10%	Please consult study summary.	(Gattefossé data, GAT-PMIC/DIF/DR 0401, 2004)
Percutaneous absorption (OECD 428)	*In vitro*	Human skin	C_{14}-TRANSCUTOL® Emulsion O/W: 2, 5, 10%	Please consult study summary.	(Gattefossé data, GAT-PMIC/DIF/DR 0401, 2004)
Percutaneous absorption (OECD 428)	*In vitro*	Human skin	C_{14}-TRANSCUTOL® Hydro-Alcoholic Gel: 15%	Please consult study summary.	(Gattefossé data, GAT-PMIC/DIF/DR 0401, 2004)
Acute irritation (JORF)	Dermal	Rabbit	TRANSCUTOL® 50% in water	Non irritant Cutaneous irritation index: 0.00	(Gattefossé data, GAT-7472, 1974)
Acute irritation (Patch test)	Dermal	Human	TRANSCUTOL® Pure	Well Tolerated Cutaneous irritation index: 0.10	(Gattefossé data, GAT-9269, 1992)
Acute irritation (JORF)	Ocular	Rabbit	TRANSCUTOL® 30% in water	Slightly Irritant Ocular irritation index (acute): 2.0 Ocular irritation index (48h): 0.0	(Gattefossé data, GAT-96334, 1996)
Acute irritation (OECD 405)	Ocular	Rabbit	TRANSCUTOL® Pure	Slightly Irritant	(Gattefossé data, GAT-96335, 1996)
Sensitisation (Human repeated insult patch test - HRIPT)	Dermal	Human	TRANSCUTOL® Pure	Non Irritant Non Sensitizing	(Gattefossé data, GAT-9370, 1993)
Local tolerance after parenteral injections	IM	Rabbit	TRANSCUTOL® 30% in olive oil	Moderately Irritant	(Gattefossé data, GAT-7972, 1979)
Local tolerance after parenteral injections	IM	Rabbit	TRANSCUTOL® 50% in water	Moderately Irritant	(Gattefossé data, GAT-7972, 1979)
Local tolerance after parenteral injections	IM	Rabbit	TRANSCUTOL® HP	not classified as irritant	(Gattefossé data, 43160TAL, 2015)
Local Tolerance Study in the Rabbit Following Intramuscular Injection	IM	Rabbit	TRANSCUTOL® HP	well-tolerated	(Gattefosse data, XC28WS, 2018)
Acute toxicity (OECD 401)	Oral (gavage)	Rat	TRANSCUTOL® Pure (undiluled) Dose levels: 5000 mg/kg	$LD50_{(oral)}$ > 5000 mg/kg	(Gattefossé data, GAT-94205, 1994)

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900

SAFETY STUDIES PERFORMED BY GATTEFOSSÉ					
Study type/ duration	**Route**	**Species**	**Test article**	**Results/Conclusions**	**Reference**
Acute toxicity (Dose escalating)	Oral (gavage)	Dog	TRANSCUTOL® HP Pure (undiluled) Dose levels: 500, 1000, 1500, 2000 mg/kg	$MTD_{(oral)}$ > 2000 mg/kg	(Gattefossé data, GAT-WIL 261016, 2007)
Subchronic 7-day toxicity study (DRF)	Oral (gavage)	Dog	TRANSCUTOL® HP Dose levels: 500, 1000, 1500, 2000 mg/kg	$MTD_{(oral)}$ > 2000 mg/kg	(Gattefossé data, GAT-WIL 261016, 2007)
Subchronic 13-week toxicity study (OECD 408)	Oral (Gavage)	Dog	TRANSCUTOL® HP Dose levels: 0, 400, 1000, 2000/1500 mg/kg/day	NOAEL (oral): 1000 mg/kg/day	(Gattefossé data, GAT- WIL261017, 2007)
Acute toxicity (Dose escalating)	IV bolus (tail vein)	Mouse	TRANSCUTOL® HP Vehicle: Physiological saline solution M: 25, 50, 100, 200, 400, 800, 1600, 6400, 3200 and 4800 mg/kg F: 25, 50, 100, 200, 400, 800, 1600, 8000, 6400, 4800 and 3200 mg/kg	$MTD_{(IV)}$: 3200 mg/kg	(Gattefossé data, GAT-MTDs-PH-08/0477, 2009)
Acute toxicity (eq OECD 423)	IV bolus (tail vein)	Mouse	TRANSCUTOL® HP Vehicle: Physiological saline solution M: 125 mg/kg F: 250 and 125 mg/kg	$MTD_{(IV)}$: 3200 mg/kg Administration at 3200 mg/kg induces no mortality and toxicity signs.	(Gattefossé data, GAT-TAIs-PH-08/0477, 2009)
Subchronic 28-day toxicity study (eq OECD 410)	Dermal	Rabbit	TRANSCUTOL® Dose levels: 0, 100, 300, 1000 mg/kg/day	$NOAEL_{(dermal)}$ > 1000 mg/kg/day Non Toxic for dose levels up to 1000 mg/kg/day	(Gattefossé data, GAT- 95241, 1995)
Fertility study (Segment I) (eq OECD 415)	Oral (Gavage)	Rat	TRANSCUTOL® HP Vehicle/control: sterile water Dose levels: 0, 300, 1000, 2000 mg/kg/day	$NOAEL_{(oral)}$: 2000 mg/kg/day No effects on gonodal function, fertility and reproductive performance for dose levels up to 2000 mg/mg/day	(Gattefossé data, GAT-MDS 935/105, 2001)
Embryofetal development study (Segment II) (eq OECD 414)	Oral (Gavage)	Rat	TRANSCUTOL® HP Vehicle/control: sterile water Dose levels: 0, 300, 1000, 2000 mg/kg/day	$NOAEL_{(oral,\ mat)}$: 1000 mg/kg/day No indication of a teratogenicity for dose levels up to 2000 mg/kg/day	(Gattefossé data, GAT-MDS 935/122, 2002)
Ames test (With and without activation) (OECD 471)	In vitro	S. typhimurium	TRANSCUTOL® P Pure	Negative. Not mutagenic.	(Gattefossé data, GAT-99493, 1999)
DNA damage and repair (UDS: Unscheduled DNA synthesis in mammali) (eq OECD 486)	In vivo/In vitro	Rat	TRANSCUTOL® Pure	Not Genotoxic for dose levels up to 2000 mg/kg/day	(Gattefossé data, GAT-96340, 1996)
Chromosome aberration (OECD 473)	In vitro	human lymphocytes	TRANSCUTOL® HP Pure	did not induce chromosome aberrations	(Gattefossé data, 43155MLH, 2016)
Bone marrow micronucleus test (OECD 474)	Intravenous	Rat	TRANSCUTOL® HP Pure	did not induce damage to the chromosomes or the mitotic apparatus up to 1500 mg/kg/day	Gattefossé data, 43156MAR 2016)
OECD : Organisation for Economic Co-operation and development - JORF: Journal official République Française (Official Journal French Republic)					
Summary of files available upon request, please contact your Gattefossé local representative. Additional literature review is available for the different endpoints, please consult the document "Tox and safety overview" for a complete review on regulatory, tox and safety aspects.					

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900

CIR COMPENDIUM (Cosmetic Ingredient Review) (http://www.cir-safety.org)			
Denomination	**Reference**	**Maximum "as used" concentration**	**Conclusions**
Butylene glycol, Hexylene Glycol, Ethoxydiglycol and Dipropylene Glycol	JACT 4(5):223-48 1985	Up to 50%	Based on the available data, Butylene glycol, Hexylene Glycol, Ethoxydiglycol and Dipropylene Glycol are safe as presently used in cosmetics.

9. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
INCI NAME	**CAS #**	**EC #**	**Chemical name**
Ethoxydiglycol	111-90-0	203-919-7	2-(2-ethoxyethoxy)ethanol

NATIONAL INVENTORIES					
INCI NAME	**U.S.A. (TSCA)**	**Canada (DSL/NDSL)**	**Australia (AIIC)**	**China (IECSC)**	**Other inventories**
Ethoxydiglycol	Listed	DSL listed	Listed	Listed	Ethanol, 2-(2-ethoxyethoxy)- (AIIC, DSL, ENCS, INSQ, NZIoC, PICCS, TCSI, TSCA, IECSC) Diethylene glycol monoethyl ether (IECSC, PICCS) 2-(2-Ethoxyethoxy)ethanol (AREC, DSL, ECL, EINECS, PICCS, REACH, VNECI) 2-(2-Ethoxyethoxy)-ethanol (TDCA)

U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics… are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting.

Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation.

Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities.
Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration.
Chemicals not listed on the AIIC must be assessed before importation.

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
	Status of the product	**Status per substance**
Ethoxydiglycol	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACh regulation.

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900

10. HISTORY OF USE

10.1. Pharmaceutical applications

Refer to the HISTORY OF USE document.

- FDA IID reference exactly match the definition of the excipient TRANSCUTOL P: Diethylene glycol monoethyl ether

10.1.1. Other references

HANDBOOK OF PHARMACEUTICAL EXCIPIENTS (Current edition by R. C. Rowe, P.J. Sheskey, W. G. Cook, M. E Fenton)			
Denomination	**Functional Category**	**Regulatory status**	**Safety**
Diethylene glycol monoethyl ether	Solubilizing agent Solvent	Included in the FDA Inactive Ingredients Database (topical and transdermal gels). Approved in the US as an indirect food additive. Included in nonparenteral formulations available in the UK and France. Included in the Canadian Natural Health Products Ingredients Database.	Diethylene glycol monoethyl ether is used in a wide variety of pharmaceutical formulations for oral, dermal, and parenteral administration, and as a cosmetic ingredient. It is generally regarded as nonirritant and nontoxic. The Cosmetic Ingredient Review (CIR) Expert Panel has evaluated diethylene glycol monoethyl ether and found it to be safe as presently used in cosmetics at a maximum concentration of 50%. The acute toxicity of diethylene glycol monoethyl ether after oral, IV, and dermal application as well as inhalation has been studied and can be regarded as very low in all species investigated. The LD50 values for acute oral, acute IV, and acute dermal toxicity were found to be generally much higher than 2 g/kg body weight, and the available LC50 value for acute inhalation was > 5 mg/L (i.e. 5.24 mg/L) in rat, in each case above the limit doses for classification. LD50 (rat, oral) > 5 g/kg LD50 (mouse, IV): 3.2 g/kg

10.2. Food/Nutraceutical/Dietary supplement applications

Diethylene glycol monoethyl ether has a history of use as indirect food additives for use in food, nutritional and dietary supplements.

Country	Reference	Registration identifier
International	Evaluations Performed by Joint FAO/WHO Expert Committee on Food Additives	TRS 859-JECFA 44/9 Diethylene glycol monoethyl ether
U.S.A.	USFA US Food additives according to 21 CFR	Indirect food additives 21 CFR § 175.105 21 CFR § 176.180 21 CFR § 178.1010

NB: The Food additive details are given for information. Gattefossé does not promote the use of this ingredient for Nutraceutical applications (food additive monograph conformity is not guaranteed). The suitability of the ingredient for use in nutraceutical products and dietary supplements remains the own responsibility of the company marketing the finished product.

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900

Liability clause

GATTEFOSSÉ SAS
36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - FRANCE
Tel +33 4 72 22 98 00 - Fax +33 4 78 90 45 67 - www.gattefosse.com
SAS au capital de 4 650 000 Euros - R.C.S. Lyon B 389 586 900 - N° d'identité CEE FR 49 389 586 900